

Member Task Assignment & Reporting Guide: Quinez, John Benedict D.

Role in Project: Lead Programmer / Software Engineer & Systems Architect

Primary Chapters: Chapter 1, Chapter 3, Chapter 4, Chapter 5

✦ Assigned Sections Checklist

Chapter 1: Introduction

- ☐ 1.1 Project Context
- ☐ 1.2 Objectives
 - ☐ 1.2.1 General Objective
 - ☐ 1.2.2 Specific Objectives (All 12 Modules):
 - ☐ 1. Standard Role-Playing Game (RPG) Combat Mechanics
 - ☐ 2. Math Battle System Plugin
 - ☐ 3. Level-Based Difficulty System
 - ☐ 4. "Content Aware" Timer System
 - ☐ 5. Enemy Auto Scaling System
 - ☐ 6. Performance Based Reward Mechanism
 - ☐ 7. Automatic Quest Generation System
 - ☐ 8. Peer to Peer (P2P) Multiplayer Connection
 - ☐ 9. Saving System
 - ☐ 10. Level Based Progression System
 - ☐ 11. Mobile Input System (Virtual Numeric Keypad)
 - ☐ 12. Game Assets and Entities

Chapter 3: Technical Background

- ☐ 3.1.3 Peopleware
 - ☐ A. The Proponents (Role distribution & collaboration)
 - ☐ B. Capstone Adviser (Mr. Jan Nicole B. Apostol)
- ☐ 3.1.4 Network
 - ☐ A. Local Area Network (LAN) & Cloud Staging Setup

Chapter 4: Methodology, Results and Discussion

- ☐ 4.2.1 Operational Feasibility
 - ☐ Functional Decomposition Diagram (FDD Structure & Hierarchy)
- ☐ 4.2.5 Requirements Modeling (4.2.5.1 Object Modeling)
 - ☐ Use Case Diagrams (Player Combat, Exploration, Quests)
 - ☐ Sequence Diagrams (Auto Quest, Enemy Scaling, Level Progression, Math Battle, P2P Multiplayer, Save System)
 - ☐ Activity Diagrams (Title Screen, Main Menu/Save, Enemy Scaling, Math Battle, Level Progression, Quest Gen, P2P Multiplayer)
- ☐ 4.4 Development Specifications & Technical Environment

- ☐ 4.4.2 Hardware Specification (Workstation Specifications: Dell OptiPlex 3010, Core i3, 16GB RAM)
- ☐ 4.4.3 Program Specification (Event-driven specifications for all 11 core modules)
- ☐ 4.4.5 Test Plan (Multi-tier testing strategy)
- ☐ **4.5 Verification, Validation, Testing**
 - ☐ 4.5.1 Unit Testing (Execution results across all modules)
 - ☐ 4.5.2.3 Stress Testing (25 concurrent user web asset simulation)
 - ☐ 4.5.2.4 Load Testing (Serverless / P2P load testing exemption justification)
- ☐ **4.6 Acceptance Testing**
 - ☐ 4.6 Acceptance Testing Overview
 - ☐ 4.6.1 Alpha Testing (Developer simulation, boundary checks, input latency)
 - ☐ 4.6.2 Beta Testing (User testing execution with Grade 4–6 students)

Chapter 5: Conclusion and Recommendations

- ☐ **5.1 Conclusion**
 - ☐ Content Evaluation (100% Comprehension rate)
 - ☐ Module Satisfaction (Auto-Scaling 100%, Quest Gen 100%, Multiplayer 100%, Math Battle 80–95%)
 - ☐ Multimedia Elements (Visuals, Font, Audio 94%–100%)
 - ☐ Navigation Usability (Menu flow & transitions 90%–95%)
 - ☐ Usefulness (100% Math utility rating for practicing mental calculations)

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In Chapter 1, you have to report:

- 1 The Core Problem & Context (1.1):** Explain why traditional educational math drills cause Mathematics Anxiety in Grade 4–6 students (aged 9–12). Highlight the "*Chocolate-Covered Broccoli*" effect (where educational games hide standard tests behind superficial graphics without integrating learning into the core gameplay loop). Explain how *Chronicles of Arithmos* solves this by making math calculations the direct driver of combat actions and damage calculations.
- 2 The Project Objectives (1.2):** Present the General Objective (to design and develop *Chronicles of Arithmos* with Dynamic Math Combat and Adaptive Scaling). Walk through the **12 Specific Objectives** sequentially, showing how each module satisfies the SMART criteria (What + How + Result).

In Chapter 3, you have to report:

- 1 Peopleware & Network Collaboration (3.1.3 & 3.1.4):** Describe the proponent team composition and the technical guidance provided by Capstone Adviser Mr. Jan Nicole B. Apostol. Explain how LAN and Cloud staging environments were configured to test WebRTC P2P signaling without relying on a centralized host server.

In Chapter 4, you have to report:

- 1 System Modeling & Diagrams (4.2.1 & 4.2.5):** Present the **Functional Decomposition Diagram (FDD)** to show the complete breakdown of game sub-systems. Walk the panel through the **UML Use Case, Sequence, and Activity Diagrams** to demonstrate how data and control flow during math battles, enemy auto-scaling, and P2P party connections.
- 2 Hardware & Program Specifications (4.4.2 & 4.4.3):** Detail the development workstation hardware and the exact event-driven tables for the 11 modules (Events triggered, state checks, action outputs).

3 Testing Execution & Results (4.5 & 4.6):

- Report the **Unit Testing** pass rates.
- Explain the **Stress Testing** findings (simulating 25 concurrent users fetching web assets from Hostinger).
- Defend the **Load Testing Exemption** (explaining that because single-player is 100% offline and multiplayer is decentralized P2P via WebRTC, no centralized backend database exists to load-test).
- Present the **Alpha & Beta Testing** methodology and user acceptance outcomes.

In Chapter 5, you have to report:

- 1 **Conclusions (5.1):** Map survey outcomes 1-to-1 against the initial specific objectives. Emphasize that 100% of student respondents confirmed the game's utility for mental math speed, and explain how the module satisfaction ratings prove the success of the combat-to-curriculum mechanics.